

CSE 2nd Year Odd Semester Syllabus

Computer Science & Engineering • Rajshahi University of Engineering & Technology

CSE 2100

Software Development Project I

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Students will develop one or more Programs/Projects on some practical problems with sound Software Engineering practices as assigned by the teacher.

CSE 2101

Discrete Mathematics

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Logic and Proof: Propositional Logic, Applications of Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Introduction to Proofs, Proof Methods and Strategy.

Set: Using Set Notation with Quantifiers, Truth Sets and Quantifiers, Generalized Unions and Intersections, Computer Representation of set.

Function: One-to-One and Onto Functions, Inverse Functions and Compositions of Functions, The Graphs of Functions, floor and ceiling function with their properties, Partial Functions.

Sequences and Summations: Sequences, recurrence relations and Summations.

Cardinality of Sets: Countable Sets, Uncountable Set, Uncomputable Functions.

Number Theory: Divisibility and Modular Arithmetic, Primes and Greatest Common Divisors, Solving Congruences, Applications of Congruences.

Induction and Recursion: Mathematical Induction, Strong Induction and Well-Ordering, Recursive Definitions and Structural Induction.

Counting: The Pigeonhole Principle, Binomial Coefficients and Identities, Generalized Permutations and Combinations, Generating Permutations and Combinations. Applications of Recurrence Relations, Generating Functions, Inclusion–Exclusion, Applications of Inclusion–Exclusion.

Relation: Property of relation, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings.

Graphs: Graphs and Graph Models, Graph Terminology and Special Types of Graphs, Representing Graphs, Graph Traversing, Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Planar Graphs, Graph Coloring.

CSE 2102

Discrete Mathematics Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Discrete Mathematics: Logic and Proof, Set, Function, Sequences and Summations, Cardinality of Sets, Relation, Number Theory, Induction and Recursion, Counting, Graphs.

CSE 2103

Digital Logic Design

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Information and Digital Systems: Introduction to Digital Systems, Number Systems, Weighted and Non-Weighted Codes, Error Detection Code, Binary Addition and Subtraction, 2's Complement Methods.

Boolean Algebra and Combinational Logic Circuits: Digital Logic, Boolean Algebra, Boolean Function, Canonical Forms, Karnaugh Maps, Minimization of Boolean Functions, Logic Gates and their Truth Tables, Design Methodologies, Combinational Logic Circuit Design, Arithmetic and Data Handling Logic Circuits. Decoders, Encoders, Multiplexer, Demultiplexer.

Flip Flop and Sequential Logic Circuits: Transistor Latch, NAND Gate Latch, NOR Gate Latch, D Latch. Clock Signals and Clocked FFs: Clocked SR, JK and D Flip-Flops, Master/Slave JK FF, Timing Diagram of Different FFs, Edge-Triggered and Level-Triggered Timing Diagrams, Counters, Registers, Memory Devices and their Applications.

Hardware Description Language (HDL): Synthesis of digital circuits using Hardware Description Language (HDL). Reference Language: Verilog.

Technology Parameters: Fan In, Fan Out, Propagation Delay, Power Dissipation and Noise Immunity.

Others: Diode Logic Gates, Transistor Gates, MOS Gates, Logic Families: TTL and CMOS Logic with Operation Details.

CSE 2104

Digital Logic Design Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Digital Logic Design: Designing Various Logic Combinational circuits, Designing Various Sequential Circuits, Hardware Description Language (HDL), Designing and Implementing Various TTL & CMOS Logic Devices. Reference HDL: Verilog.

EEE 2151

Electrical Drives and Instrumentations

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

DC Machines: Operation and Performance Characteristics of Generators and Motors, Starting, Speed Control and Braking of Motors, Different Application of DC Motors.

AC Machines: Transformer: Principle of Operation of Transformer, Single Phase Induction Motors and its Methods of Starting, Introduction to alternator.

Special Machines: Stepper Motor, Servomotor, Brush less Motors.

Measuring Instruments: Electromechanical and Electronic Meters and their Uses, Extension of Instrument Range.

Instrumentation Amplifiers: Differential, Logarithmic and Chopper Amplifiers, Frequency and Voltage Measurements using Digital Techniques, Recorders and Display Devices, Spectrum Analyzers and Logic Analyzers.

Transducers: Terminology, Types, Principles and Application of Photovoltaic, Piezoelectric, Thermoelectric, Position and Displacement Transducers, Potentiometer, Linear Variable Differential Transformers (LVDT), Pressure Transducer, Temperature Transducer, Optical Transducer, Ultrasonic Transducer, Humidity Transducer, Hall Effect Transducer, Speed Transducer, Noise Reduction in Instrumentation.

EEE 2152

Electrical Drives and Instrumentations Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Electrical Drives and Instrumentation: Transformer, DC Machine, AC Machine, Measuring Instruments and Transducers.

Math 2113

Vector Analysis and Linear Algebra

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Vector Analysis: Different Types of Vectors, Vector Differentiation and Integration, Line, Surface and Volume Integrals, Gradient of a Function, Divergence and Curl of Vector and Their Applications, Physical Significance of Gradient, Divergence and Curl, Vector Identities, Integral Forms of Gradient, Divergence and Curl, Green's Theorem, Stokes' Theorem, Gauss's Divergence Theorem.

Matrix: Definition of Matrices, Equality of two Matrices, Addition, Subtraction and Multiplication of Matrices, Equivalence of Matrices, Positive and Negative Matrices, Adjoint of Matrices, Transpose and Inverse of Matrices, Rank and Normal Form of Matrices, System of Linear Equations, Solution of Homogeneous and Non-Homogeneous Systems, Determination of Eigen Values and Eigen Vectors, Solution of Matrix Differential Equations.

Linear Algebra: Vector Space, Subspace, Sum and Direct Sum, Hilbert Space, Normed Linear Space, Branch Space, Basis and Dimension, Linear Transformation: Range, Kernel, Nullity, Singular and Non-Singular transformation, Linear Operations: Matrix Representation of a Linear Operator, Change of Basis, Similarity and Linear Mapping.

Hum 2113

Economics, Government and Sociology

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Economics: Introduction of Economics, Demand Supply and Their Elasticity, Theory of Production, Theory of Cost, Economic Development and Economic Planning, Concept of National Income, Economic Policies.

Government: Institutions: State - Forms & Functions of State, Government-Forms & Functions of Government, Local & Central Government, Judiciary & Legislative System, Political, Economic & Recreational Institutions and their Roles in the Society, Formal Organization & Bureaucracy, Public Opinion.

Sociology: Introduction: Define Society, Evaluation of Society, distinguish between Community and Society, Culture, Define Sociology, Importance of Sociology, Scope of Sociology. Social System: Family, Marriage, Role of Engineers in the Society, Effects of Change in Technology on Society, Ethical Perspectives of Technology. Culture: Elements of Culture, Cultural Systems and Subsystems, Cultural Lag, Relationship between Culture and Civilization. Social Problems: Unemployed Delinquency and Crime, Social Crime, Cyber Crime, Deviant Behavior, Globalization and Human Rights. Others: Geographical Location of Bangladesh, Urban Community, Urban Ecology, Rural & Urban Power Structure.
